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Reporte de instalación.

Sistema operativo: Ubuntu

Versión: 24.04 LTS

Distribución: Xubuntu.

Versión de la instalación(de Docker): 29.7.2

Tiempo requerido: 20 minutos.

Procedimiento

Instalar Docker y PostgreSQL.

1. Actualizamos algunos paquetes.
2. Se instalan paquetes esenciales para el repositorio docker.

```
Removing runc (1.3.3-0ubuntu1-24.04.3) ...
Processing triggers for man-db (2.12.0-4build2) ...
(Reading database ... 466292 files and directories currently installed.)
Purging configuration files for ubuntu-fan (0.12.16+24.04.1) ...
Purging configuration files for bridge-utils (1.7.1-1ubuntu2) ...
Purging configuration files for containerd (1.7.28-0ubuntu1-24.04.2) ...
s-hu3ro2@ubuntu:~$ sudo apt install ca-certificates curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl3t64-gnutls libcurl4t64
The following packages will be upgraded:
  ca-certificates curl libcurl3t64-gnutls libcurl4t64
4 upgraded, 0 newly installed, 0 to remove and 475 not upgraded.
Need to get 0 B/1.044 kB of archives.
After this operation, 51.2 kB disk space will be freed.
Do you want to continue? [Y/n] Y
Preconfiguring packages ...
(Reading database ... 466292 files and directories currently installed.)
Preparing to unpack .../ca-certificates_20240601-24.04.1_all.deb ...
Unpacking ca-certificates (20240601-24.04.1) over (20240203) ...
Preparing to unpack .../curl_8.5.0-2ubuntu10.13_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.13) over (8.5.0-2ubuntu10.9) ...
Preparing to unpack .../libcurl4t64_8.5.0-2ubuntu10.13_amd64.deb ...
Unpacking libcurl4t64 (8.5.0-2ubuntu10.13) over (8.5.0-2ubuntu10.9) ...
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.13_amd64.deb ...
Unpacking libcurl3t64-gnutls (8.5.0-2ubuntu10.13) over (8.5.0-2ubuntu10.9) ...
Setting up libcurl4t64:amd64 (8.5.0-2ubuntu10.13) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.13) ...
Setting up ca-certificates (20240601-24.04.1) ...
Updating certificates in /etc/ssl/certs...
rehash: warning: skipping ca-certificates.crt, it does not contain exactly one certificate or CRL
14 added, 39 removed; done.
Setting up curl (8.5.0-2ubuntu10.13) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu7) ...
Processing triggers for ca-certificates (20240601-24.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Processing triggers for ca-certificates-java (20240118) ...
done.
s-hu3ro2@ubuntu:~$
```

3. Creamos un directorio donde se almacenan llaves de los repositorios, descargamos la llave GPG, damos permisos de lectura y agregamos el repositorio oficial de docker.

```
done.
s-hu3ro2@ubuntu:~$ sudo install -m 0755 -d /etc/apt/keyrings-
s-hu3ro2@ubuntu:~$ sudo curl -fSSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
s-hu3ro2@ubuntu:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
s-hu3ro2@ubuntu:~$ sudo tee /etc/apt/source.list.d/
docker.sources <<EOF
Types: deb
URIs:
https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo $
{UBUNTU_CODENAME:-SVERSION_CODENAME})
Components: stable
Architectures: $(dpkg
--print-architecture)
Signed-By: /etc/apt/keyrings/
docker.asc
EOF
tee: /etc/apt/source.list.d/: Is a directory
echo$
{UBUNTU_CODENAME:-noble}; command not found
dpkg: error: need an action option

Type dpkg --help for help about installing and deinstalling packages [*];
Use 'apt' or 'aptitude' for user-friendly package management;
Type dpkg --get-selections for a list of packages;
Type dpkg --get-architecture for a list of architectures;
Type dpkg --force-help for a list of forcing options;
Type dpkg-deb --help for help about manipulating *.deb files;

Options marked [*] produce a lot of output - pipe it through 'less' or 'more' !
--print-architecture: command not found
docker.sources: command not found
s-hu3ro2@ubuntu:~$
```

- Actualizamos nuevamente nuestros paquetes.
- Instalamos docker.

```
File Edit View Terminal Tabs Help
Terminal - s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA-
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras pigz
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin pigz
0 upgraded, 7 newly installed, 0 to remove and 479 not upgraded.
Need to get 102 MB of archives.
After this operation, 395 MB of additional disk space will be used.
Do you want to continue? [5/n] 5
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 2.3.4-1-ubuntu.24.04-noble [22,7 MB]
Get:2 https://archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65,6 kB]
Get:3 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:29.7.2-1-ubuntu.24.04-noble [17,0 MB]
Get:4 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:29.7.2-1-ubuntu.24.04-noble [24,0 MB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.36.1-1-ubuntu.24.04-noble [17,2 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:29.7.2-1-ubuntu.24.04-noble [10,2 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-compose-plugin amd64 5.5.0-1-ubuntu.24.04-noble [11,1 MB]
Fetched 102 MB in 32s (3.240 KB/s)
Selecting previously unselected package containerd.io.
(Reading database ... 460257 files and directories currently installed.)
Preparing to unpack .../0-containerd.io_2.3.4-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking containerd.io (2.3.4-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5:29.7.2-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-cli (5:29.7.2-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5:29.7.2-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce (5:29.7.2-1-ubuntu.24.04-noble) ...
Selecting previously unselected package pigz.
Preparing to unpack .../3-pigz_2.8-1_amd64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../4-docker-buildx-plugin_0.36.1-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-buildx-plugin (0.36.1-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../5-docker-ce-rootless-extras_5:29.7.2-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-rootless-extras (5:29.7.2-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-compose-plugin.
Preparing to unpack .../6-docker-compose-plugin_5.5.0-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-compose-plugin (5.5.0-1-ubuntu.24.04-noble) ...
Setting up containerd.io (2.3.4-1-ubuntu.24.04-noble) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.service.
Setting up docker-compose-plugin (5.5.0-1-ubuntu.24.04-noble) ...
Setting up docker-ce-cli (5:29.7.2-1-ubuntu.24.04-noble) ...
Setting up pigz (2.8-1) ...
Setting up docker-ce-rootless-extras (5:29.7.2-1-ubuntu.24.04-noble) ...
Setting up docker-ce (5:29.7.2-1-ubuntu.24.04-noble) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Could not execute systemctl: at /usr/bin/deb-systemd-invoke line 148.
Processing triggers for man-db (2.12.0-4ubuntu2) ...
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$
```

- Comprobamos que este funcionando.

```
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ sudo systemctl status docker
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Tue 2026-09-01 11:46:25 CST; 3min 8s ago
 TriggeredBy: x docker.socket
   Docs: https://docs.docker.com
   Process: 13599 ExecStart=/usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock (code=exited, status=1/FAILURE)
 Main PID: 13599 (code=exited, status=1/FAILURE)
    CPU: 48ms

sep 01 11:46:25 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA systemd[1]: docker.service: Scheduled restart job, restart counter is at 3.
sep 01 11:46:25 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA systemd[1]: docker.service: Start request repeated too quickly.
sep 01 11:46:25 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA systemd[1]: docker.service: Failed with result 'exit-code'.
sep 01 11:46:25 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA systemd[1]: Failed to start docker.service - Docker Application Container Engine.
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ sudo rm -f /etc/docker/daemon.json
rm: invalid option -- '/'
Try 'rm --help' for more information.
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ sudo rm -f /etc/docker/daemon.json
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ sudo systemctl start docker
sudo: systemctl: command not found
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ sudo systemctl start docker
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ sudo systemctl status docker
s-hu3r02@s-hu3r02-Zenbook-15-UM3504DA-UM3504DA:~$ docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-01 11:57:49 CST; 35s ago
 TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
   Main PID: 14828 (dockerd)
    Tasks: 17
  Memory: 29.3M (peak: 31.3M)
     CPU: 546ms
    CGroup: /system.slice/docker.service
            └─14828 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA dockerd[14828]: time="2026-09-01T11:57:49.081223471-06:00" level=info msg="Restoring containers: start."
sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA dockerd[14828]: time="2026-09-01T11:57:49.103986019-06:00" level=info msg="Deleting nftables IPv4 rules" error="running nft: /dev/stdin:1:18-30: Error: Could not process rule: No such rule"
sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA dockerd[14828]: time="2026-09-01T11:57:49.130491395-06:00" level=info msg="Deleting nftables IPv6 rules" error="running nft: /dev/stdin:1:18-31: Error: Could not process rule: No such rule"
sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA dockerd[14828]: time="2026-09-01T11:57:49.657488718-06:00" level=info msg="Loading containers: done."
sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA dockerd[14828]: time="2026-09-01T11:57:49.664520946-06:00" level=info msg="Docker daemon commit=6a43e3d containerd-snapshotter=true storage-driver=overlays version=29.7.2"
sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA dockerd[14828]: time="2026-09-01T11:57:49.664642366-06:00" level=info msg="Initializing buildkit"
sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA dockerd[14828]: time="2026-09-01T11:57:49.870785220-06:00" level=info msg="Completed buildkit initialization"
sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA dockerd[14828]: time="2026-09-01T11:57:49.875783969-06:00" level=info msg="Daemon has completed initialization"
sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA dockerd[14828]: time="2026-09-01T11:57:49.875836710-06:00" level=info msg="API listen on /run/docker.sock"
sep 01 11:57:49 s-hu3r02-Zenbook-15-UM3504DA-UM3504DA systemd[1]: Started docker.service - Docker Application Container Engine.
Lines 1-22/22 [END]
```

7. Descargamos la imagen de postgres y creamos un contenedor con esa imagen.

```
File Edit View Terminal Tab Help
Terminal - s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA: ~

Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA:~$ docker pull postgres-
invalid reference format
s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA:~$ docker pull postgres
Using default tag: latest
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA:~$ sudo docker pull postgres
Using default tag: latest
latest: Pulling from library/postgres
1b554e877727: Pull complete
4e6d3ebcb0f5: Pull complete
a4635eeec9b8: Pull complete
6463cb5c703: Pull complete
a6eb7177594: Pull complete
5fa29c524b4e: Pull complete
6310eb1b0f42: Pull complete
b6e4226a207d: Pull complete
2fc0e69f8d7d: Pull complete
af2a6480a058: Pull complete
56b94bff1337: Pull complete
ace4b5b06a91: Pull complete
5ee6cca77f09: Pull complete
6ff88415323f: Download complete
091cb76c0e1d: Download complete
Digest: sha256:4ef4dbc939d61acea57712655ddb4b27419c913f94cca0cd57cb3ea3c2280
Status: Downloaded newer image for postgres:latest
docker.io/library/postgres:latest
s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA:~$ docker run -d \
--name postgres \
-e POSTGRES_PASSWORD=mysecretpassword \
-p 127.0.0.1:5432:5432 \
-v postgres_data:/var/lib/postgresql \
postgres
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA:~$ sudo docker run -d --name postgres -e POSTGRES_PASSWORD=mysecretpassword -p 127.0.0.1:5432:5432 -v postgres_data:/var/lib/postgresql postgres
```

8. Revisamos el id del contenedor recién creado.

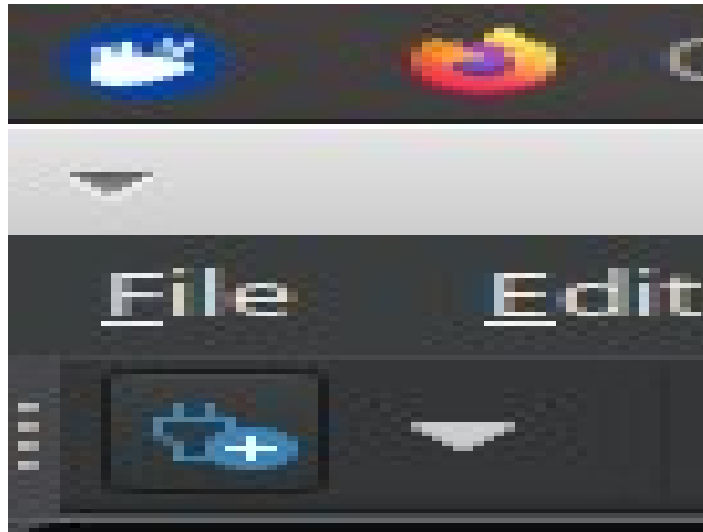
9. Probamos iniciar el contenedor.

```
s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA:~$ sudo docker start 13be093f1f3d
13be093f1f3d
s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA:~$ sudo docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
WARNING: Links on the default bridge network are deprecated and will be removed in a future release. Use a custom network instead.
Password for user postgres:
psql: error: connection to server at "postgres" (172.17.0.2), port 5432 failed: FATAL: password authentication failed for user "postgres"
s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA:~$ sudo docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
WARNING: Links on the default bridge network are deprecated and will be removed in a future release. Use a custom network instead.
Password for user postgres:
psql (18.6 (Debian 18.6-1.pgdg13+2))
Type "help" for help.

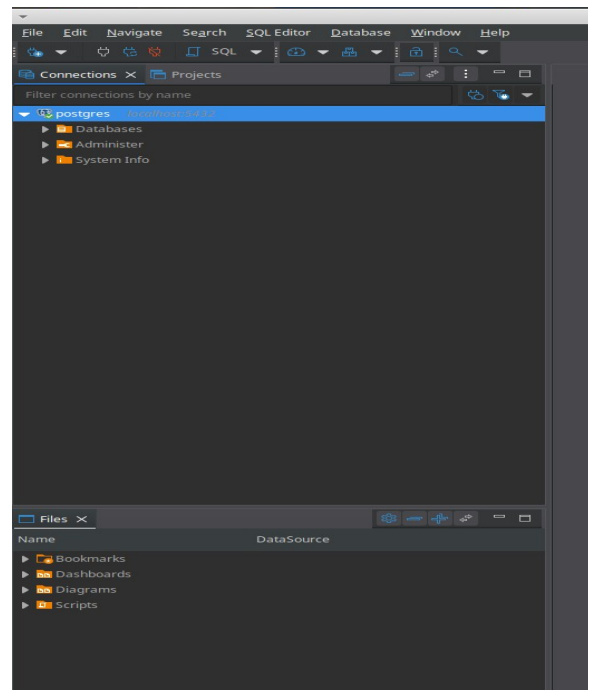
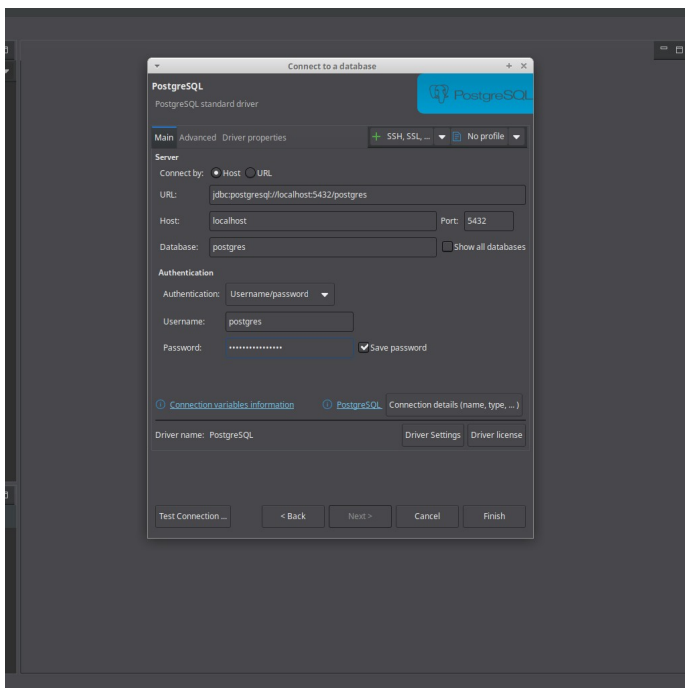
postgres=# /q
postgres=# \q
s-hu3ro2@hu3ro2-Zenbook-15-UM3504DA-UM3504DA:~$
```

Vincular Dbeaver

1. Abrimos Dbeaver y seleccionamos la opción "New Database Connection".



2. En "Select your Database" seleccionamos PostgreSQL.
3. Escribimos la contraseña y listo.



Problemas y comentarios

En el paso número 6 sucedió que docker daba un error y decía que no estaba activo. Para solucionar este error, se usaron los comandos `sudo systemctl start docker` y `sudo systemctl enable docker` para activar el servicio de contenedores y hacer que docker se inicie automáticamente respectivamente.

Esta información se obtuvo del siguiente tutorial:

<https://youtu.be/jX5UYznr5Xs?si=xvSe> utilizó el siguiente tutorial: ra9XLp74-R53CI

Por lo demás, la mayoría del tiempo solo fue copiar y pegar los comandos, aunque como se puede ver en las fotos a veces los escribía sin el sudo . Como dato adicional, ya después de terminar todo, agregue mi usuario a docker para no tener que usar sudo (me apoyé de lo visto en clase).

